

Roton-like Reconstruction of the Emergent Photon in Quantum Spin Ice

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The emergent photon of quantum spin ice is well understood at long wavelength, but at lattice momenta the compact $U(1)$ theory has no small parameter—the electric field is saturated at $S^z = \pm \frac{1}{2}$, so a field wave cannot be reduced in amplitude, only rearranged—and which eigenstates carry the spin response there has never been resolved. We compute the operator-resolved $S^{zz}(\mathbf{q}, \omega)$ of the pure pyrochlore ring-exchange model on 48-, 64-, 72-, and 96-site clusters, the largest a constrained sector of dimension 1.47×10^8 . In the Gaussian theory the two transverse modes are exactly degenerate at every momentum, and no renormalization of its parameters can split them; we measure a splitting that grows with $|\mathbf{q}|$ and reaches a factor 1.7 at the zone-boundary L point, where the lower member collapses to half the Gaussian energy and takes 50–54% of the longitudinal weight, against 0.82–1.11 of that energy at every other momentum resolved. The dominant zone-boundary response is therefore not a photon. That mode is elementary—the lowest excitation of its sector—and its photon-like partner is excluded from being a two-mode composite by spin-reversal parity and by kinematics independently. Its softness is predicted before any excited state is computed by the Feynman ratio $f_1(\mathbf{q})/S(\mathbf{q})$, a rigorous variational bound built from ground-state data, which ranks all five resolved momenta in exactly the order of their measured softening; its two factors identify the mechanism as an exact curl cancellation on one hexagon family multiplied by a quantum enhancement of the static correlations that is absent in the classical ice manifold. The reconstruction is stable across four clusters, three cluster shapes, and a factor 1.6×10^4 in Hilbert-space dimension, and sign-free Green’s-function Monte Carlo carries the ground-state diagnostic to 2048 spins with its minimum still at L . The phenomenology is Feynman’s and the object is a lattice roton, predicting a two-feature zone-boundary response whose energy ratio 0.55–0.59 and two-to-one weight ratio are both independent of the overall energy scale.

Introduction.—Quantum spin ice realizes a compact $U(1)$ lattice gauge theory in a magnetic insulator. In the two-in-two-out manifold of the pyrochlore lattice, virtual exchange resonates the six spins around an elementary hexagon, and the resulting deconfined phase supports a gapless emergent photon together with gapped electric and magnetic charges [1–6]. That long-wavelength description is quantitatively established and motivates the search for candidate materials [7–16].

At lattice momenta the same theory is a different problem, and the reason is structural. The emergent electric field is microscopically saturated, $E_\ell = S_\ell^z = \pm \frac{1}{2}$, so $\sum_\ell E_\ell^2$ is a constant inside the constrained manifold: there is no electric stiffness to tune, no coupling constant to send to zero, and hence no free-field limit from which the interacting theory can be reached perturbatively. The emergent fine-structure constant is large [17]. A field wave cannot be made small in amplitude, only rearranged in position—which is a statement about short-wavelength physics, precisely the regime the Gaussian expansion does not control.

Existing numerics stop short of that regime in a specific sense. Gaussian and semiclassical treatments expand about small flux and are controlled in the infrared, though they already predict substantial anharmonic renormalization and an interaction-induced splitting of the transverse polarizations [7, 18–20]. Quantum Monte Carlo with analytic continuation resolves broadened lineshapes and finds zone-boundary weight beyond

Gaussian electrodynamics, but not individual eigenstates or their matrix elements [21]. Constrained-space exact diagonalization has reached 96 spins, organized around ground-state and low-energy electrodynamic parameters rather than the full response [17]. What has not been done is the direct question: at large $|\mathbf{q}|$, *which* eigenstates of the compact theory carry the spin response, and how much.

We answer it. Our punchlines are:

1. In the Gaussian theory the two transverse modes are *exactly* degenerate at every momentum of our clusters, and any renormalization of its two parameters rescales both equally. We measure splittings up to a factor 1.69 at the zone-boundary L point, and they are not a function of $|\mathbf{q}|$: two momenta at identical $|\mathbf{q}|$ split by 1.69 and 1.06. This is a qualitative failure of the Gaussian description, not a quantitative correction to it, and it is momentum-specific rather than a generic large- $|\mathbf{q}|$ effect.
2. At L the lower member does not merely split off; it collapses to 0.50 of the Gaussian energy and takes 50–54% of the longitudinal weight. At every other momentum we resolve it sits between 0.82 and 1.11 of the Gaussian energy. The dominant zone-boundary response is therefore not a photon.
3. This mode is elementary—the lowest excitation of its sector at every size where the low-lying spectrum is available—and its photon-like partner cannot be

a two-mode composite, by spin-reversal parity and by kinematics independently.

4. Its softness is quantitatively predicted, before any excited state is computed, by the Feynman ratio $f_1(\mathbf{q})/S(\mathbf{q})$: this ground-state quantity is a rigorous variational upper bound, and it ranks all five resolved momenta in exactly the order of their measured softening. The mechanism it exposes is an exact lattice cancellation in the oscillator strength multiplied by a quantum enhancement of the static correlations.
5. The reconstruction is stable across four clusters, three cluster shapes, and a factor 1.6×10^4 in Hilbert-space dimension, and sign-free Monte Carlo carries the ground-state diagnostic to 2048 spins. It predicts a two-feature zone-boundary response whose energy ratio and weight ratio are both independent of the overall energy scale.

Model, observable, and method.—We study the standard ring Hamiltonian in the spin-ice manifold [1, 8], $H = -K \sum_p (W_p + W_p^\dagger)$ with $W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-$ summed over contractible elementary hexagons; in gauge language $W_p = e^{iB_p}$ and H is the compact magnetic energy $-2K \sum_p \cos B_p$. Because S^z is the emergent electric field, the natural probe is the sublattice-summed longitudinal response $S^{zz}(\mathbf{q}, \omega) = N^{-1} \sum_{\mu, n} |\langle n | A_{\mathbf{q}}^\mu | 0 \rangle|^2 \delta(\omega - \omega_n)$ with $A_{\mathbf{q}}^\mu = \sum_{i \in \mu} e^{-i\mathbf{q} \cdot \mathbf{r}_i} S_i^z$. All spectral fractions below are normalized within S^{zz} at fixed $\mathbf{q}$. This is the theoretical pseudospin response, not a material-specific neutron cross section, which additionally involves coherent sublattice interference, the local easy axes, the g tensor, form factors, and the polarization projector; we treat that fully in the Supplemental Material.

We evaluate S^{zz} by complete diagonalization at 48 sites and by Lanczos continued fractions at 64, 72, and 96, the last in a winding sector of dimension 146 758 352. Winding sectors are certified exhaustively at 48 and 72 sites—all 155 and 310 sectors diagonalized, margins $0.99K$ —and by the twelve largest of 519 at 96, where the margin is $0.375K$. Recomputing the 96-site spectra at Krylov dimension 200 rather than 120 leaves the dominant pole energies and residues unchanged to machine precision. Details are in the Supplemental Material.

The Gaussian doublet and what replaces it.—The Gaussian theory of this model is lattice electrodynamics with two parameters, an electric stiffness U and the ring coupling K . Diagonalizing the lattice curl at momentum $\mathbf{q}$ gives its transverse spectrum, $\omega_\lambda(\mathbf{q}) = \sqrt{2UK} \sigma_\lambda(\mathbf{q})$, with σ_λ the nonzero singular values of the curl. On every cluster and at every momentum we examined, exactly two are nonzero and they are *equal* to machine precision: $\sigma_1 = \sigma_2$, e.g. $2\sqrt{3}$ at L and 4 at X . The two transverse polarizations are degenerate throughout the zone, and because U and K enter only through the prefactor, that

degeneracy survives *any* renormalization of the Gaussian theory, however strong. A splitting of the doublet cannot be produced within that description at all.

Figure 1 shows what the exact spectrum does instead. At the smallest momentum we resolve, the two dominant levels are nearly degenerate—their ratio is 0.957—and both sit within 12% of the Gaussian energy, obtained by fixing the single scale $\sqrt{2UK}$ from the long-wavelength spectrum. This is a weakly renormalized photon doublet, as expected. At the remaining momenta the doublet splits, by factors from 1.06 to 1.69.

The splitting is momentum-specific, not a generic consequence of large $|\mathbf{q}|$, and the cluster provides its own control. At $|\mathbf{q}| = 0.866$ exactly, L splits by 1.69 while $(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$ splits by only 1.06; at $|\mathbf{q}| = 1$ exactly, X splits by 1.53 and $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ by 1.19. Identical wavevector magnitude, splittings differing by up to a factor of 1.6: whatever produces the effect selects particular momenta, not a shell.

Splitting alone, however, is still not the phenomenon. At X the doublet splits substantially yet its lower member sits at 0.952 of the Gaussian energy—split, but nothing has softened. At L both happen at once: the splitting is largest *and* the lower member falls to 0.503 of the Gaussian energy, half. It is that combination, not the splitting by itself, that we call the reconstruction, and L is the only momentum in Table I where it occurs. The softening is likewise not ordered by $|\mathbf{q}|$ —the same two momenta at $|\mathbf{q}| = 0.866$ give 0.503 and 1.114.

The mode is elementary.—Before interpreting it we establish what it is not. At 48, 64, and 72 sites, where the low-lying spectrum is available in full, the reconstructed level is the *lowest excitation of its sector*— $1.024K$, $0.954K$, $1.068K$, with nothing between it and the ground state. It cannot be a composite of anything.

Its photon-like partner cannot be a two-mode composite either, and two independent arguments exclude it. Global spin reversal $S^z \rightarrow -S^z$ is a symmetry of H ; the ground state is even and $A_{\mathbf{q}}$ is odd, so only odd states carry weight in S^{zz} . Any state built from an even number of elementary excitations is even and therefore exactly

TABLE I. All momenta resolved at 48 sites, ordered by the ground-state predictor. ω_G is the Gaussian energy at the single fitted scale; the Gaussian doublet is exactly degenerate at every entry. The predictor orders the five momenta exactly as the measured softening $\omega_{\text{low}}/\omega_G$ does.

$\mathbf{q}$	$S(\mathbf{q})$	f_1/S	$(f_1/S)/\omega_G$	$\omega_{\text{low}}/\omega_G$	$\omega_{\text{low}}/\omega_{\text{high}}$
L	0.370	1.846	0.908	0.503	0.592
$(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$	0.287	2.491	1.177	0.824	0.843
X	0.273	3.223	1.373	0.952	0.653
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	0.189	2.420	1.375	1.076	0.957
$(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$	0.258	3.216	1.415	1.114	0.943

dark to this probe—we verify the complementarity numerically to 10^{-25} . The upper branch carries 22–32% of the weight, so it is odd. Independently, kinematics: two elementary excitations cost at least $2 \times 1.024K = 2.047K$ at 48 sites and three cost $3.07K$, while the upper branch lies at $1.730K$, below the two-particle threshold. Both branches are single elementary excitations.

Why L : the oscillator strength.—The location of the reconstruction is not an accident of the cluster, and it can be derived. The two lowest moments of S^{zz} are ground-state quantities. The zeroth is the equal-time structure factor, $m_0 = \int d\omega S^{zz} = \langle A_{\mathbf{q}}^\dagger A_{\mathbf{q}} \rangle \equiv S(\mathbf{q})$, which measures how much static correlation at wavevector $\mathbf{q}$ the ground state already holds. The first follows from the f -sum rule,

$$m_1 = \int d\omega \omega S^{zz}(\mathbf{q}, \omega) = \frac{1}{2} \langle [A_{\mathbf{q}}^\dagger, [H, A_{\mathbf{q}}]] \rangle. \quad (1)$$

For the ring Hamiltonian this evaluates in closed form. Since W_p raises S^z on alternating sites of its hexagon and lowers it on the others, $[A_{\mathbf{q}}, W_p] = g_p(\mathbf{q})W_p$ with $g_p(\mathbf{q}) = \sum_{j \in p} (-1)^j e^{-i\mathbf{q} \cdot \mathbf{r}_j}$, the *alternating lattice curl* of the imposed field wave around p . Then $[H, A_{\mathbf{q}}] = K \sum_p g_p(W_p - W_p^\dagger)$, and Eq. (1) collapses to

$$f_1(\mathbf{q}) \equiv m_1 = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu |g_p^\mu(\mathbf{q})|^2. \quad (2)$$

This has the form of a Maxwell magnetic energy and reads directly: the cost of imposing an electric-field wave of wavevector $\mathbf{q}$ is the squared curl that wave drives through each plaquette, weighted by how strongly that plaquette actually resonates in the ground state. It contains no fitted parameter and no excited state.

Their ratio,

$$\omega_F(\mathbf{q}) = \frac{f_1(\mathbf{q})}{S(\mathbf{q})} = \frac{\langle 0 | A_{\mathbf{q}}^\dagger (H - E_0) A_{\mathbf{q}} | 0 \rangle}{\langle 0 | A_{\mathbf{q}}^\dagger A_{\mathbf{q}} | 0 \rangle}, \quad (3)$$

is simultaneously two things. Trivially it is the centroid of $S^{zz}(\mathbf{q}, \omega)$, first moment over zeroth. More usefully, the right-hand side of Eq. (3) is the expectation value of $H - E_0$ in the normalized single-mode state $A_{\mathbf{q}}|0\rangle/\sqrt{S(\mathbf{q})}$, which is orthogonal to the ground state for $\mathbf{q} \neq 0$. Hence ω_F is a *rigorous variational upper bound* on the lowest odd excitation at $\mathbf{q}$. This is Feynman’s construction for the helium roton [22], and it is why the quantity is worth computing: a small ω_F does not describe a soft mode after the fact, it *certifies* from ground-state data alone that one exists.

It does more than certify. Normalized to the Gaussian energy so that momenta of different $|\mathbf{q}|$ are comparable, ω_F/ω_G ranks all five resolved momenta in exactly the order of their measured softening (Table I), with no inversions: L is predicted softest and is softest, $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ second and is second, and the three momenta the predictor places together at 1.37–1.42 are the three that do not

soften at all. Since neither the softening nor the splitting is ordered by $|\mathbf{q}|$, this is not a trivial correlation: on the shell $|\mathbf{q}| = 0.866$ the predictor assigns 0.908 to L and 1.415 to $(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$, and the measured softenings are 0.503 and 1.114. The bound is not tight— $\omega_F(L) = 1.846K$ against a true $1.024K$, since the centroid also carries the high-energy weight—but as an ordering of which momenta reconstruct it is exact on this data.

Equation (2) also says why L wins, and it takes both factors. Geometrically, for $\mathbf{q} \parallel [111]$ one of the four hexagon families lies entirely within the wavefronts of the field oscillation, so its alternating curl vanishes identically: the family-resolved geometric weights are $(96, 96, 0, 96)$ at L against $(96, 96, 96, 96)$ at X . A quarter of the plaquettes cost nothing and f_1 drops. Geometry alone is not enough—the momentum $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ has a family cancellation too, $(72, 0, 72, 72)$, and does not soften, because its S is the smallest in the table. The second ingredient is dynamical: $S(L) = 0.370$ is the largest, while the equal-amplitude superposition over the same ice configurations gives only 0.297 and classical loop Monte Carlo on a 2048-site manifold is flat among zone-boundary momenta. The enhancement is made by the interacting quantum amplitudes, not by the ice constraint—which is what distinguishes this from a Rokhsar–Kivelson resonon, whose softening follows from classical correlations. L is the one momentum where a vanishing curl family coincides with the strongest static correlations, and Eq. (3) multiplies the two.

Interpretation: this is a roton.—We use the term in Feynman’s specific sense—the collective mode created by the density-wave operator at finite momentum, soft because f_1/S has a minimum there—and not as a synonym for “low-lying mode.” Four measured statements identify the reconstructed branch as that object, none of which refer to it as “the lower of two levels.”

First, it is what the density-wave operator makes: the single-mode state $A_{\mathbf{q}_L}|0\rangle$ overlaps the reconstructed multiplet with squared amplitude 0.88, 0.87, and 0.86 at 48, 64, and 72 sites, coupling to its partner only through the orthogonal transverse channel. The Feynman construction produces this branch and not the other; that, rather than an energy ordering, is the categorization. Second, it is elementary, as established above. Third, its softness is predicted by ω_F in rank order across momenta. Fourth, relaxing the single-mode state within the exact spectrum recovers 28–32% of its energy, a size-stable amount, through a strongly family-selective reorganization of ring coherence—the correlated relaxation that in helium carries Feynman’s bound down toward the true roton.

The lattice modifies the geometry of the phenomenon and we state that plainly. Along the radial $[111]$ direction the centroid rises from Γ to L and folds back by periodicity, so there is no helium-like radial minimum; the minimum lives in the zone-boundary manifold, and

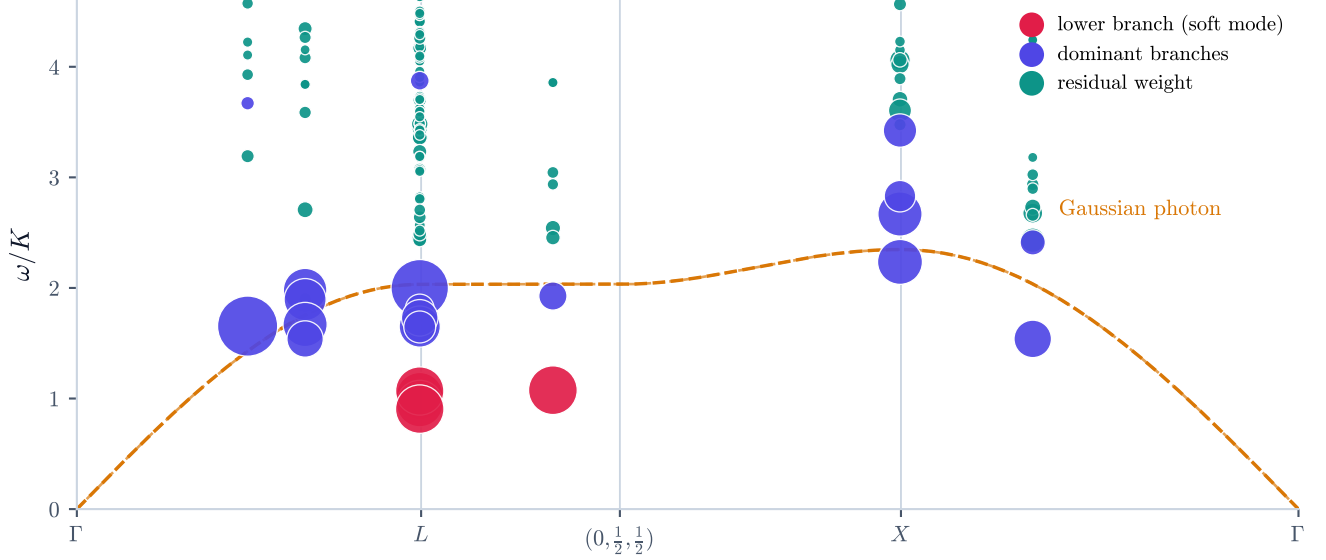


FIG. 1. **Order-one reconstruction at lattice momentum.** Operator-resolved S^z spectrum along $\Gamma \rightarrow L \rightarrow (0, \frac{1}{2}, \frac{1}{2}) \rightarrow X \rightarrow \Gamma$, combining all four clusters; marker area is the fraction of the sublattice-summed weight carried by each level. The dashed curve is the Gaussian photon at a single fitted scale, whose two transverse polarizations are exactly degenerate everywhere. Near Γ the two dominant levels track it and each other. Near L the doublet splits by a factor 1.69 and its lower member collapses to half the Gaussian energy while absorbing half the response.

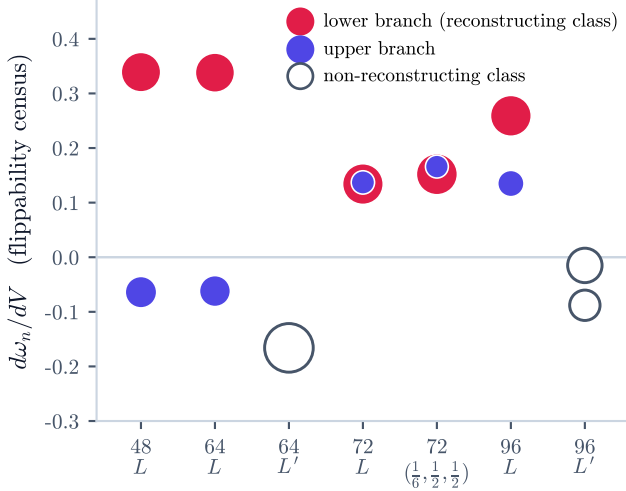


FIG. 2. **Flippability census.** State-resolved $d\omega_n/dV$ from Eq. (4). The reconstructed mode gains flippable hexagons at every size and momentum where it occurs (filled), whereas every dominant level at a momentum that does not reconstruct loses them (open). The sign is a property of the momentum, not of a branch ordering: at 72 and 96 sites the photon-like partner at a reconstructing momentum is also positive.

its shallow extension along $(t, \frac{1}{2}, \frac{1}{2})$ is why the 72-site cluster finds a second reconstructed momentum on that line.

What the mode does microscopically.—One further property uses no fitted scale and no Gaussian compar-

ison at all. Adding the Rokhsar–Kivelson term $H(V) = H + V \sum_p n_p^{\text{flip}}$, with n_p^{flip} the projector onto a flippable alternating hexagon, and applying Hellmann–Feynman at the physical point $V = 0$ gives

$$\left. \frac{d\omega_n}{dV} \right|_{V=0} = \left\langle n \left| \sum_p n_p^{\text{flip}} \right| n \right\rangle - \left\langle 0 \left| \sum_p n_p^{\text{flip}} \right| 0 \right\rangle, \quad (4)$$

an exact state-resolved census of the flippable hexagons an excitation adds or removes. Flippability is the resource that ring resonance consumes: a hexagon must be flippable to contribute to the energy at all. The ground state balances having many flippable plaquettes against phasing them coherently, and a typical excitation loses on both counts—averaged over a sector the census is exactly $\text{nnz}(H)/D - \langle n^{\text{flip}} \rangle_0 < 0$, and at 48 sites only 58 of 9104 eigenstates are positive.

The reconstructed mode is one of them, at every size and every momentum where the reconstruction occurs: +0.339, +0.338, +0.134, +0.152, +0.259. It is resource-rich and coherence-poor—it rearranges spins so that *more* hexagons could flip while the ring processes on them are less coherent, paying its energy in resonance rather than in the ice rule. Conversely, every dominant level at a momentum that does *not* reconstruct loses flippability: the inequivalent L classes at identical $|\mathbf{q}|$ on the anisotropic 64- and 96-site clusters give −0.166 and −0.015, −0.088 (Fig. 2).

We are explicit about the limit of this diagnostic, because it bears on how much weight the interpretation can carry. The sign tracks the *momentum*, not the branch

ordering: at a reconstructing momentum the photon-like partner is negative at 48 and 64 sites but positive at 72 and 96, so a positive census is not by itself proof that a given level is the roton. What is uniform across all our data is the pair of statements just made—the reconstructed mode always gains, everything at a nonreconstructing momentum always loses—and it is in that form that the census enters the argument.

Robustness.—The three cluster shapes $2 \times 2 \times n$, $2 \times 3 \times 3$, and $2 \times 3 \times 4$ sample different boundary identifications and the reconstruction survives all of them (Table II, Fig. 3). The most robust measures need no fitted scale: the branch ratio is 0.55–0.59 across a factor 1.6×10^4 in sector dimension, and the weight fraction is 50–54%. Absolute energies drift by about ten percent, as expected at these sizes, and the census magnitude drifts more. Not everything is stable and we say which: the polarization splitting at X falls from 42% at 48 sites to 6% at 64, and we draw no conclusion from it.

Because ω_F needs only the ground state, it can be pushed far beyond exact diagonalization. The ring Hamiltonian is stoquastic in the ice basis, permitting sign-free Green’s-function Monte Carlo with forward walking, benchmarked against the exact 48-site $S(L)$ to every digit. On complete allowed momentum grids of 864 and 2048 spins the ratio at L is 2.142(10) and 2.179(19), against 2.660(10) and 2.713(21) at X and 2.72–2.78 at W , K , and U : an L -centered minimum 20–25% deep that drifts by 1.7% between the two sizes (Fig. 4). The diagnostic that predicts the reconstruction at 48 spins still has its minimum at L at 2048.

Experimental implications.—The signature is a two-feature zone-boundary response where a renormalized photon gives one. Following [111] outward, a single dispersing mode at small q separates into two, the splitting growing from unresolvable to a factor 1.7 by the zone boundary, with the *lower* feature carrying two to three times the weight of the upper. Two aspects are scale-free and so testable without knowing K : the ratio of the two energies, 0.55–0.59, and the ratio of their weights. A single photon branch with a renormalized velocity produces neither, and—this is the sharper point—no Gaussian theory produces a splitting at all, since its doublet is

TABLE II. The reconstruction across all four clusters, at L .

sites	48	64	72	96
sector dimension	9104	2.5×10^5	9.7×10^5	1.5×10^8
ω_{low}/K	1.024	0.954	1.068	0.904
weight	50.8%	50.0%	54.0%	54.4%
ω_{high}/K	1.730	1.652	1.814	1.647
weight	32.3%	31.3%	17.8%	22.5%
$\omega_{\text{low}}/\omega_{\text{high}}$	0.592	0.578	0.589	0.549
$d\omega_{\text{low}}/dV$	+0.339	+0.338	+0.134	+0.259

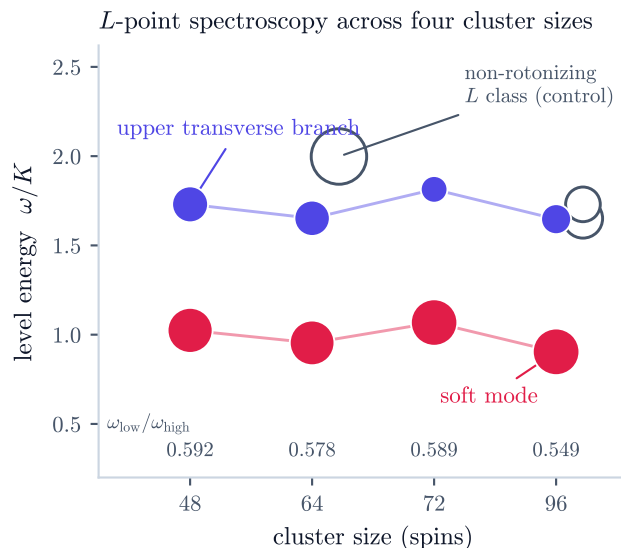


FIG. 3. **Finite-size stability and internal controls.** Filled: the reconstructed mode and its partner at L . Open: dominant levels of the inequivalent nonreconstructing class at identical $|q|$ on the anisotropic 64- and 96-site clusters. Numbers give $\omega_{\text{low}}/\omega_{\text{high}}$.

degenerate for every choice of parameters. Because the reconstruction tracks f_1/S rather than the label L , the same signature should appear along $(t, \frac{1}{2}, \frac{1}{2})$, as it does in our 72-site spectrum.

Two caveats bound this. A material-specific neutron cross section reweights the sublattice amplitudes coherently, so the theoretical 50–54% fraction is not a universal neutron intensity; and longer ring exchanges, virtual spinons, and other microscopic terms can modify the zone-boundary spectrum of a real material. The robust prediction of the minimal model is the redistribution itself.

Discussion.—The emergent photon of quantum spin ice is nonperturbatively reconstructed at the lattice scale. Its transverse doublet, exactly degenerate in the Gaussian theory for any parameters, splits by a factor 1.7 at the zone boundary, and the split-off member falls to half the Gaussian energy while taking half the longitudinal weight. That member is an elementary excitation; it is the mode the electric-field density wave creates; its softness is predicted in rank order by a ground-state bound whose two factors identify the mechanism—an exact curl cancellation on one hexagon family, multiplied by quantum-enhanced static correlations—and it is microscopically distinguished by gaining ring flippability where nonreconstructing momenta lose it. The phenomenology is Feynman’s and the object is a lattice roton.

What makes this possible is the structural point we began with: the compact theory has no small parameter at short wavelength. Because $E_\ell = \pm \frac{1}{2}$ is saturated, a finite-amplitude field wave cannot be built by increas-

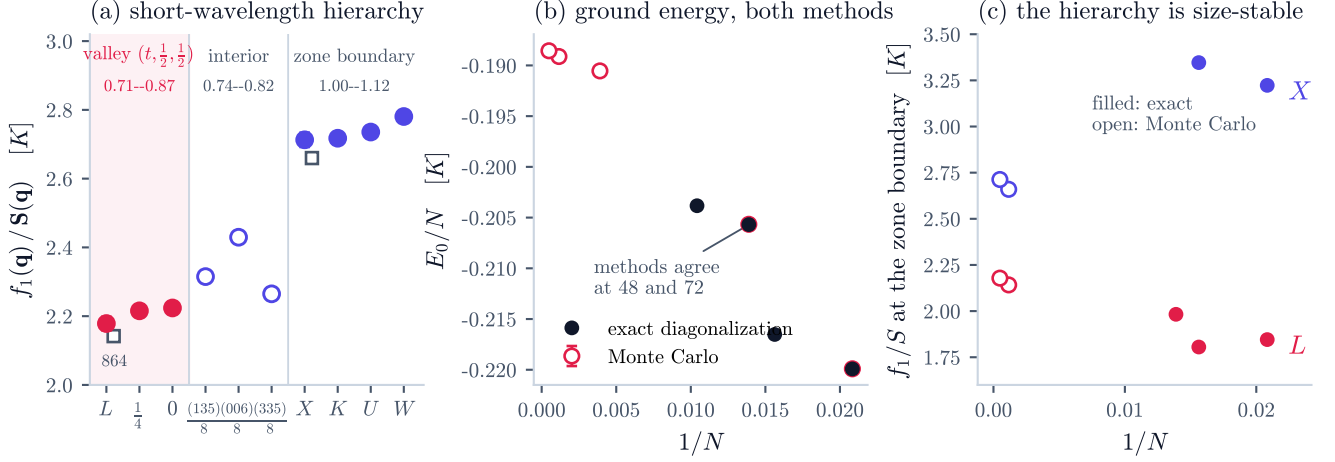


FIG. 4. **Large-system ground-state landscape.** (a) $f_1(\mathbf{q})/S(\mathbf{q})$ on the 2048-site supercell grouped by geometric role, with each group's $|\mathbf{q}|$ range indicated; at matched $|\mathbf{q}| = 0.75$ the valley sits 9–10% below the interior star $(0, 0, \frac{3}{4})$. Open squares: 864-site values at shared momenta. (b) Ground-state energy from exact diagonalization and Monte Carlo. (c) Size evolution at L and X across both methods.

ing an amplitude, only by rearranging which bonds carry which of the two allowed values—and that rearrangement is cheapest where the curl vanishes and the static correlations are strongest. A Gaussian photon has no such freedom, which is why no renormalization of it reaches this mode.

Sharpness is established up to 96 spins; beyond that the Monte Carlo constrains the bound, not the pole, and whether the level broadens into a narrow continuum in the thermodynamic limit is open. The natural next questions are whether the reconstruction survives virtual spinons and longer ring exchanges in microscopic XXZ models, and whether it can be seen directly—a question that requires dynamical rather than ground-state Monte Carlo.

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